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Failure Analysis of Acrylonitrile Butadiene Styrene (ABS) Materials and Damage Modeling by Fracture

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Abstract

Polymer materials are distinguished by their simple and economical for-mating, versatility, lightness, and chemical stability. Despite their widespread use both in everyday life and in advanced technologies, these materials are generally still very poorly understood in regards to its chemical, physical and environmental, rheological and mechanical properties. The purpose of this study is to determine the notch effect on the mechanical behavior of Acrylonitrile Butadiene Styrene (ABS), which we carried out tensile tests on ABS rectangular test pieces for different backgrounds. The notch ranges from 1 mm to 18 mm. Subsequently we have based an approach, which consists in following the evolution of static damage as a function of the fraction of life test pieces notched, to know the critical fraction of life and which corresponds to a critical notch radius, which can be authorized. Second, we calculated the stress concentration factor by two methods to determine the critical stress concentration that usually precedes the sudden failure of a component. Finally, we will relate the damage factor D_s to the stress concentration factor through the fraction of life by using a nonlinear correlation, which makes it possible to predict the instant of acceleration of damage for a preventive maintenance.

Keywords: polymer; ABS; mechanical properties; stress concentration factor; damage

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1. Introduction

Polymers have invaded industrial applications. They have become ubiquitous in all areas of application. They occupy an important class by their uses and by the diversity of the targeted applications [1]. There are a lot of researches that has been done on different types of polymers, among those researches there is the work of Ouardi [2], which is based on the determination of the critical life of PPR pipes, and also on PVC pipes which has been developed by ARID [3]. Thermoplastic polymer such as Acrylonitrile Butadiene Styrene (ABS) is a material commonly used in several sectors of activity and particularly when it is desired to combine the impact resistance with a beautiful final appearance as well as when looking for eco-design [4]; this is because ABS is a recyclable material. It is preferred for the manufacture of household appliances, vehicle interior, and hospital platforms for X ray. ABS is thermoplastic from the Styrenics family. It is a copolymer of acrylonitrile butadiene styrene invented in 1946 in the United States [5]. ABS is composed of three monomers: acrylonitrile, butadiene and styrene. It is obtained by modifying a copolymer resin (styrene acrylonitrile) using a copolymer (butadiene, acrylonitrile) [6]. Several studies are interested in the study of the mechanisms of damage and their influence on the mechanical characteristics in the mode I was the object of the works of Ni [1], noting that the shape of the plastic zone in the neighborhood of a point of stress concentration cannot be described by the criteria of Von put or Tresca. In addition, the influence of damage coupled with cyclic tests has been the subject of several studies. Gongyao et al [7] have studied the damaged behavior of thermoplastics by uniaxial tensile tests on dumbbell-shaped test pieces. The same type of behavior was invested by Malaya Ranjan et al [8] on composite hole plates subjected to multiaxial fatigue tests. In the framework of damage mechanics, researchers have developed theoretical expressions of damage such as Miner, Strakey and

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Shanly. Bui-Quoc developed the unified theory of metal fatigue and suggested damage expressions to quantify damage based on initial and current fatigue limits. In order to use this model of polymers, the characterization, through static tests, requires a judicious adaptation taking into account different parameters.

As a result, we based a static damage study to determine the critical fraction of life corresponding to a critical notch bottom, and subsequently calculated the stress concentration factor to see which bottom of the notch our material is not usable. Finally, we try to find a relation between the static damage and the stress concentration factor through the fraction of life to see the level of static damage corresponds to the value of critical stress concentration factor.

2. Material and Experimental Methods

The polymer used in this work is thermoplastic Acrylonitrile Butadiene Styrene; they are a family of copolymers for technical purposes because of their high physical and chemical properties: rigidity, impact resistance, heat resistance, surface appearance, and resistance to many chemical agents [9]. This choice of material is justified by the compatibility of ABS with a wide range of material, which makes it extremely versatile [10].

For the material characterization, uniaxial tensile tests will be performed according to ASTM D638-03 [11], ASTM D 882-02 [12] and ASTM D5766M [13], Figure 1 below shows the drawings of definition of the test specimens used for the tests.

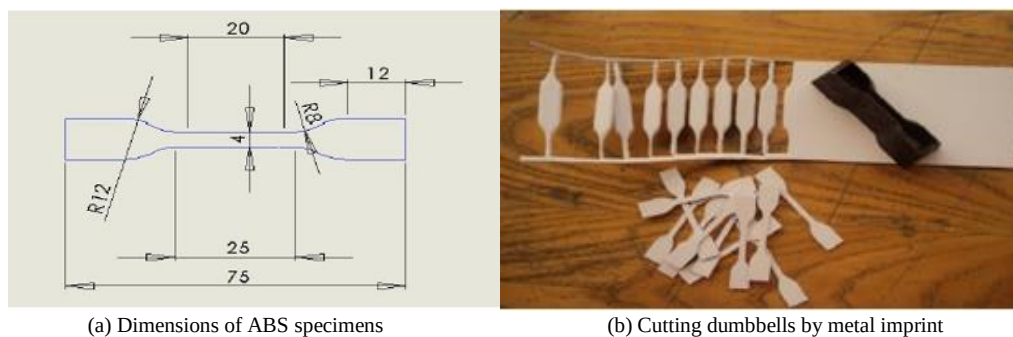


Figure 1. Specimen size according to ASTM D638-03

To study the effect of the defect on the material used (ABS), we carried out tensile tests on rectangular specimens of the ABS simply notched; according to the ASTM D 882-02 [12] and 766m ASTM [14] standards, all of these test pieces were tested in monotonous displacement with imposed displacement corresponding to a strain rate of 1mm/min. The tests are carried out under the conditions of air and ambient temperature (≈ 20 to 24°C) on a ZwickRoell type machine with a load cell of 2.5kN. Figure 2 shows the defects that we made in the form of semicircles with radius ranging from 1mm to 18mm

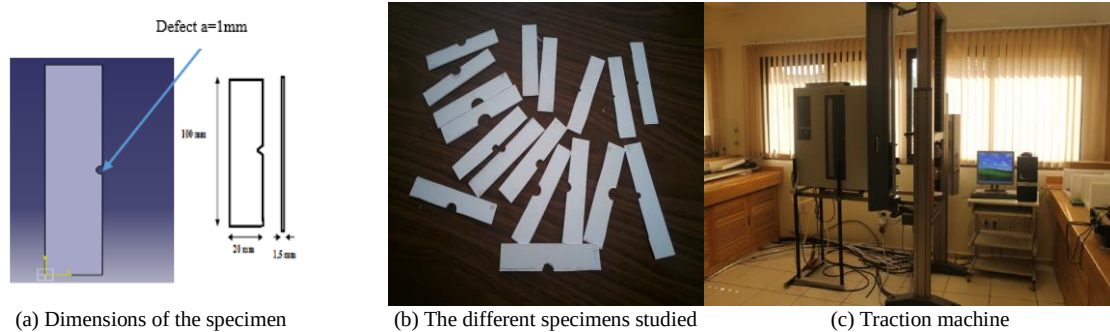


Figure 2. Specimen size according to ASTM D638-03

3. Results and Discussion

3.1. Characterization of the Material

The set of tests leading to the rupture of the ABS specimens made it possible to trace the shape of the tensile curve which represents the evolution of the stress (MPa) as a function of the strain (in %), as shown in Figure 3). Extract the mechanical characteristics and group them in a table. Table 1 shows the average values.

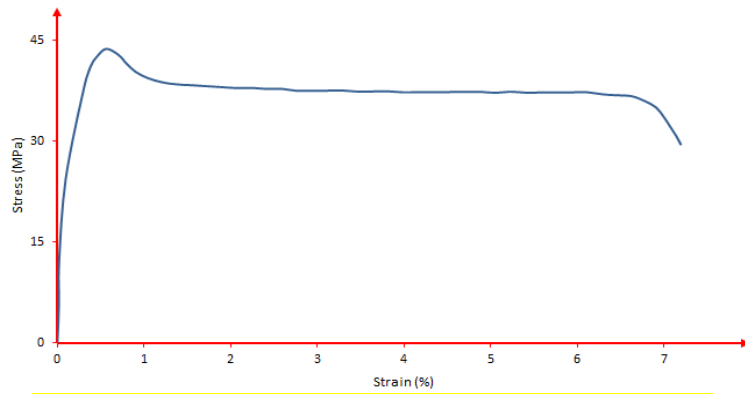


Figure 3. Evolution of the stress (MPa) according to the strain (%) of the ABS

Table 1. The mechanical properties of ABS

Mechanical properties	Values
Tensile strength	43.8MPa
Elastic limit	38.45MPa
Young modulus	1.47GPa
Poisson ratio	0.3
Elongation at break in (%)	7.2
Stress at break	29.58MPa

It can be seen from the results obtained and illustrated by the stress-strain curve that the evolution of the stress as a function of the deformation of the ABS material follows the behavior of the polymers in large deformation. The stress-strain curve has three phases. We note proportionality between the applied force and deformation; it is indeed the zone of elastic deformation where the material always has its property to return to its original shape after it has been deformed when the forces are no longer exercised. This area allows us to calculate the Young's model as well as the elastic limit. In carrying out the tests and according to the results shown in Figure 3, it has been found that the general shape of the tensile curve has demonstrated a ductile behavior.

3.2. Study of the Influence of the Defect on the Mechanical Behavior of ABS

The stress-strain curve of the tensile tests of notched specimens with different incision lengths from 1 to 18mm is illustrated in Figure 4.

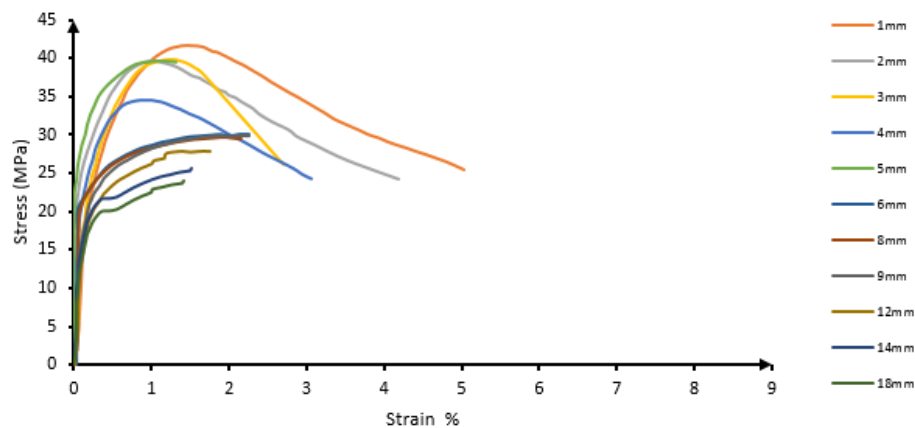


Figure 4. Strain-stress curve of notched test specimen

The evolution of the stress-strain curve as a function of the notch bottom (half hole in our case) is shown in Figure 4; the curves show increasing and decreasing trends, an offset between the different curves as a function of the bottom.

The notch is apparent. The degradation of the mechanical properties is always noticeable. The elastic stress, the ultimate stress, the tensile stress and the elongation take values that decrease with the increase of the notch bottom. We no

longer observe a zone of stabilization of the stresses or a significant elongation. The rupture often precede a local plasticization and a sudden break thereafter.

3.3. Calculates Static Damage

The static damage model consists in determining the evolution of the stress whose variations are mainly due to the damage. The static damage is then quantified by the variable D_s expressed by [15]:

$$D_s = \frac{\sigma_u - \sigma_{ur}}{\sigma_u - \sigma_a} \quad (1)$$

Where σ_u is the ultimate stress of virgin material, σ_{ur} is the residual ultimate stress, and σ_a is the stress before breaking.

In addition, the fraction of life defined as follows:

$$\beta = \frac{a}{w} \quad (2)$$

Where a is the bottom of the notch, and w is the width of the test specimen.

The parameters a and w are defined according to the geometry of our material, as shown in Figure 5.

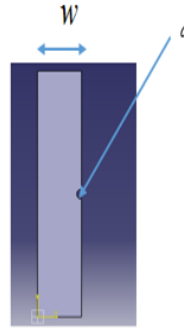


Figure 5. Parameters of test specimen

The model of static damage has been developed to assess the damage for ABS specimens. This model is based on Equation (1), through the residual stress of the undamaged and notched specimens. The damage evolution as a function of the fraction of life fraction is given in Figure 6.

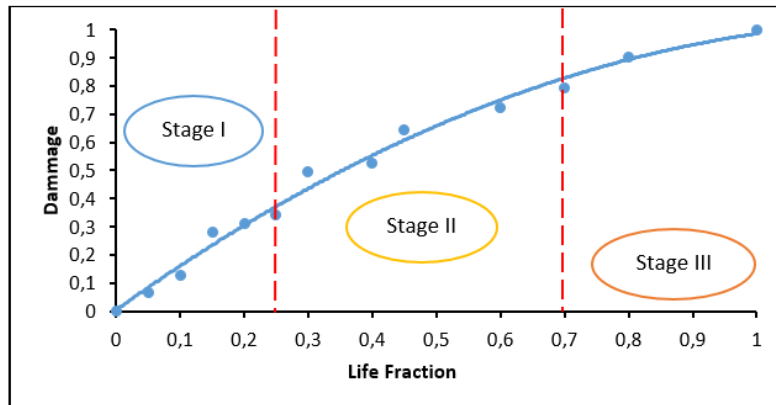


Figure 6. Static damage variation in function of the life fraction

From the curve above, it is noted that the damage D_s increases gradually with the increase of the notch. This is explained by a loss of mechanical properties caused by the depth of the notch; from the curve, one can specify the evolution of damage in three stages, according to the change of curvature. The first stage is the initiation of failure for a fraction of life ($0\% < \beta < 27\%$). The damage starts at 0 and slowly develops to 27%, which means that the specimen begins to lose its internal resistance. The second stage is the propagation of the failure. It is the phase where we reach the critical fraction of life, which corresponds to a critical radius of the notch. The fraction of life of this stage is between 27% and 70%, corresponding at a notch bottom of between 3mm and 14mm. Second stage damage ranges from 40% to 70%. At this stage, damage becomes progressive and dangerous, so predictive maintenance is essential for the industry. From $\beta = 70\%$, the third stage begins; the damage accelerates to reach a value of 1 for a completely damaged test piece. In this stage, the damage accelerates abruptly and rupture can be abrupt. This means that if $\beta = 70\%$ or greater, the test specimen becomes unusable.

3.4. Calculation of Stress Concentration Coefficient for the Test Specimen Studied

Stress concentration is a problem often encountered in the mechanical design of Acrylonitrile Butadiene Styrene. It is a phenomenon of local increase of the stresses in the zone, which involves a geometrical modification of the piece (half hole in our case). It appears in a discontinuity of the part or structure with the presence of a notch after the machining for example. The stress concentration zone is often the site of initiation of fatigue cracks but can also be the cause of a sudden rupture in the case of a fragile material. In this paper, we created discontinuities on ABS specimens in the form of half-holes (Figure 5). We varied the notches to see its effects on the stress concentration coefficient.

3.4.1. Calculation of Stress Concentration Coefficient as a Function of Notch Background a

For a given type of loading, the ratio between the actual stress (in the case where it is less than the elastic limit) and the nominal stress at a point does not depend on the value of the applied load. In this case, K_t is defined by the following relation (Equation (3)).

$$K_t = \sigma_{réelle} / \sigma_{nom} \quad (3)$$

From Equation (3) and Figure 5, actual stress and the nominal stress is defined as follows:

$$\sigma_{nom} = \frac{F}{(w-a)t} \quad (4)$$

$$\sigma_{réelle} = F / (wt) \quad (5)$$

We replace Equation (4) and Equation (5) in Equation (3):

$$K_t = \frac{w}{w-a} \quad (6)$$

Based on Equation (6), we will plot the stress concentration coefficient curve K_t as a function of the notch, which corresponds to the different test pieces studied.

According to the curve shown in Figure 7, we notice that the coefficient of stress concentration evolves progressively with the notch bottom since it is a value of $K_t=3.33$ for a notch of 14mm is the value of the critical notch a_c , and increases sharply until reaching $K_t=10$ for a cut of 18mm. These high values of the stress concentration coefficient reflect the degree of stress concentration at the level of the defect and therefore evaluate the degree of criticality in the case treated.

3.4.2. Coefficient of Concentration of Stress as a Function of Static Damage

In this section, we try to find a relation between the static damage and the coefficient of concentration of stress.

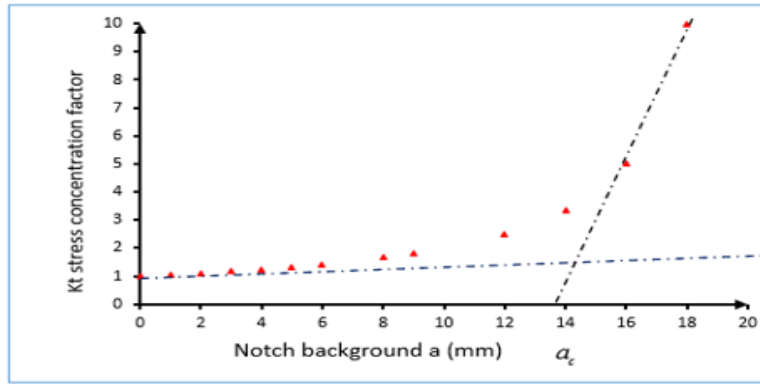


Figure 7. The evolution of stress concentration coefficient as a function of notch background

From Equation (1), we assume that:

$$\frac{\sigma_{ur}}{\sigma_u} = l e^{\lambda \beta} \quad (7)$$

Where β is the fraction of life.

The linearization of the data is as follows:

$$\ln\left(\frac{\sigma_{ur}}{\sigma_u}\right) = \ln(l) + \lambda \beta \quad (8)$$

We suppose:

$$y = \ln\left(\frac{\sigma_{ur}}{\sigma_u}\right) \quad (9)$$

$$a_0 = \ln(l)$$

$$a_1 = \lambda$$

We get:

$$y = a_0 + a_1 \beta$$

This is a linear relationship between y and β .

$$a_1 = \frac{n \sum_{i=1}^n \beta_i y_i - \sum_{i=1}^n \beta_i \sum_{i=1}^n y_i}{n \sum_{i=1}^n \beta_i^2 - \left(\sum_{i=1}^n \beta_i \right)^2}$$

$$a_0 = \bar{y} - a_1 \bar{\beta}$$

After the simplification, we found that $l = 1.005$ and $\lambda = -1.2620$; we propose Equation (10) of the damage defined by exponential:

$$D_s = c_1 + c_2 e^{\lambda \beta} \quad (10)$$

We developed Equation (1), and subsequently

$$D_s = \frac{1}{1 - \frac{\sigma_a}{\sigma_u}} - \frac{\frac{\sigma_{ur}}{\sigma_u}}{1 - \frac{\sigma_a}{\sigma_u}}$$

Then we get Equation (11):

$$D_s = \frac{1}{1 - \frac{\sigma_a}{\sigma_u}} - \frac{le^{\lambda\beta}}{1 - \frac{\sigma_a}{\sigma_u}} \quad (11)$$

By identification (Equation (12)) and (Equation (13)), we found that:

$$c_1 = \frac{1}{1 - \frac{\sigma_a}{\sigma_u}}$$

and

$$c_2 = -\frac{l}{1 - \frac{\sigma_a}{\sigma_u}}$$

From Equation (10), we obtain Equation (12) :

$$e^{\lambda\beta} = \frac{D_s - c_1}{c_2} \quad (12)$$

Moreover, later, we found that:

$$\beta = \ln\left(\frac{D_s - c_1}{c_2}\right) \times \frac{1}{\lambda} \quad (13)$$

From Equation (6), we found that:

$$\beta = \frac{K_t - 1}{K_t}$$

Then we get the following equation:

$$K_t = \frac{1}{1 - \left(\ln\left(\frac{D_s - c_1}{c_2}\right) \times \frac{1}{\lambda}\right)} \quad (14)$$

$$K_t = \frac{1}{1 - (-0,788 \ln(\frac{D_s - 1,3913}{-1,3920}))} \quad (15)$$

Equation (15) makes it possible to draw the curve of variation of the coefficient of concentration of stress as a function of the damage.

From Figure 8, we note that the stress concentration coefficient varies slowly with damage during stage I and II, and that when the damage is equal to 0.7. The coefficient of stress concentration gradually increases to reach its maximum value; this damage value coincides with a critical life fraction corresponds to a critical notch bottom β_c depending on the solicitation. This can have very important practical consequences, as far as the maintenance is concerned. This is because it makes it possible to predict the moment when it is recommended to change the part. Indeed, the theory of damage considers that the damage reaches its maximum value 1 when the material is destroyed; on the other hand, the theoretical mechanics of the linear rupture considers that K_t is equal to 1 for a smooth material and tends to infinity for a damaged material.

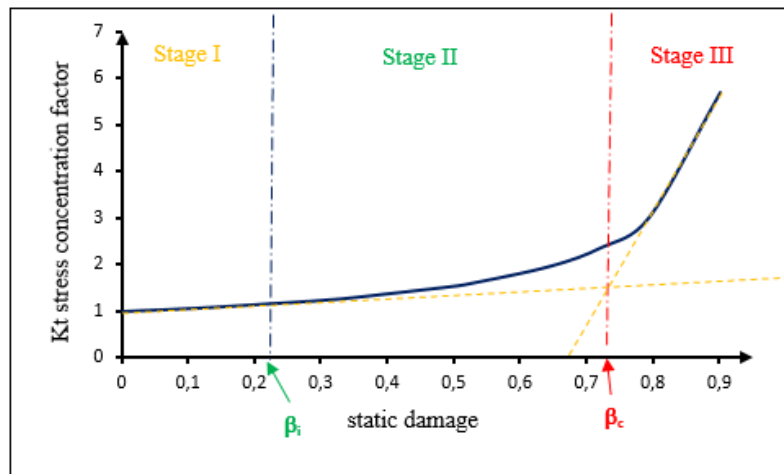


Figure 8. The evolution of stress concentration coefficient as a function of static damage D_s .

4. Conclusions

In this article, we were able to relate the stress concentration coefficient of a damaged ABS specimen to different notch lengths. We expressed the damage according to the fraction of life using the nonlinear regression methods and also found the fraction of life function of the damage by the inversion of the function of the damage. This approach led us to a new relation between mechanics of the linear rupture and the theory of damage, associating with each stage of damage the corresponding constraint concentration coefficient and vice versa. On the other hand, this approach has allowed us to predict the moment of critical damage, and therefore to intervene in time for predictive maintenance. This is especially important in advanced industry.

References

1. B. Ni, J. Li, and V. Berry, "Plastic Zone in Front of a Mode I Crack in Acrylonitrile-Butadiene-Styrene Polymers," *Polymer*, Vol. 32, No.15, pp. 2766, 1991
2. A. Ouadi, F. Majid, N. Mouhib, and M. Elghorba, "Residual Life Prediction of Defected Polypropylene Random Copolymer pipes (PPR)," *Frattura ed Integrità Strutturale*, Vol. 12, No. 43, pp. 97-105, 2018
3. G. Arid, I. Makadir, A. Naji, M. Chergui, and M. ElGhorba, "Comparative Study of the Mechanical Behavior of Polymer Materials Between ABS and PVC," *The International Journal of Engineering and Sciences*, Vol. 4, No. 4, pp. 55-60, 2015
4. H. Essabir, R. Bouhfid, and A. el Kacem Qaiss, "Fracture Surface Morphologies in Understanding of Composite Structural Behavior," *Structural Health Monitoring of Biocomposites, Fibre-Reinforced Composites and Hybrid Composites*, pp. 277, Elsevier, 2019
5. D. Ghosh and D. Khastgir, "Degradation and Stability of Polymeric High-Voltage Insulators and Prediction of Their Service Life Through Environmental and 14 Accelerated Aging Processes," *ACS Omega*, Vol. 3, No. 9, pp. 11317, 2018
6. F. Carrasco, O. Santana, J. Cailloux, M. Sanchez-Soto, and M. L. Maspoch, "Poly (Lactic Acid) and Acrylonitrile- Butadiene-Styrene Blends: Influence of Adding ABS-g-MAH Compatibilizer on the Kinetics of the Thermal Degradation," *Polymer Testing*, Vol. 67, pp. 468, 2018
7. G. Gu, Y. Xia, C. -H. Lin, S. Lin, Y. Meng, and Q. Zhou, "Experimental Study on Characterizing Damage Behavior of Thermoplastics," *Materials and Design*, Vol. 44, pp. 199, 2013
8. M. R. Satapathy, B. Vinayak, K. Jayaprakash, and N. Naik, "Fatigue Behavior of Laminated Composites with a Circular Hole under in-Plane Multiaxial Loading," *Materials and Design*, Vol. 51, pp. 347, 2013
9. D. -J. Liaw, B. -Y. Liaw, and C. -M. Yang, "Synthesis and Properties of New Polyamides based on Bis [4-(4-Aminophenoxy) Phenyl] Diphenylmethane," *Macromolecules*, Vol. 32, No. 21, pp. 7248, 1999
10. H. Farid, K. Elhad, M. Elghorba, F. Erchiqui, and M. Chergui, "Damage Characterization of a Thin Plate Made of ABS under Uniaxial Solicitation," *International Journal of Engineering and Science*, Vol. 2, No. 8, pp. 11-17, 2013

11. ASTM D638-14, Standard Test Method for Tensile Properties of Plastics, ASTM International, 2015
12. ASTM D882-12, Standard Test Method for Tensile Properties of Thin Plastic Sheeting, ASTM International, 2012
13. ASTM d5766/d5766m-07-standard test method for open hole tensile strength of polymer matrix composite laminates ,2007
14. R. M. OHiggins, M. A. McCarthy, and C. T. McCarthy, "Comparison of Open Hole Tension Characteristics of High Strength Glass and Carbon Fibre-Reinforced Composite Materials," *Composites Science and Technology*, Vol. 68, No. 13, pp. 2770, 2008
15. A. Wahid, N. Mouhib, A. Kartouni, H. Chakir, and M. ELghorba, "Energy Method for Experimental Life Prediction of Central Core Strand Constituting a Steel Wire Rope," *Engineering Failure Analysis*, doi: doi.org/10.1016/j.engfailanal.2018.12.005